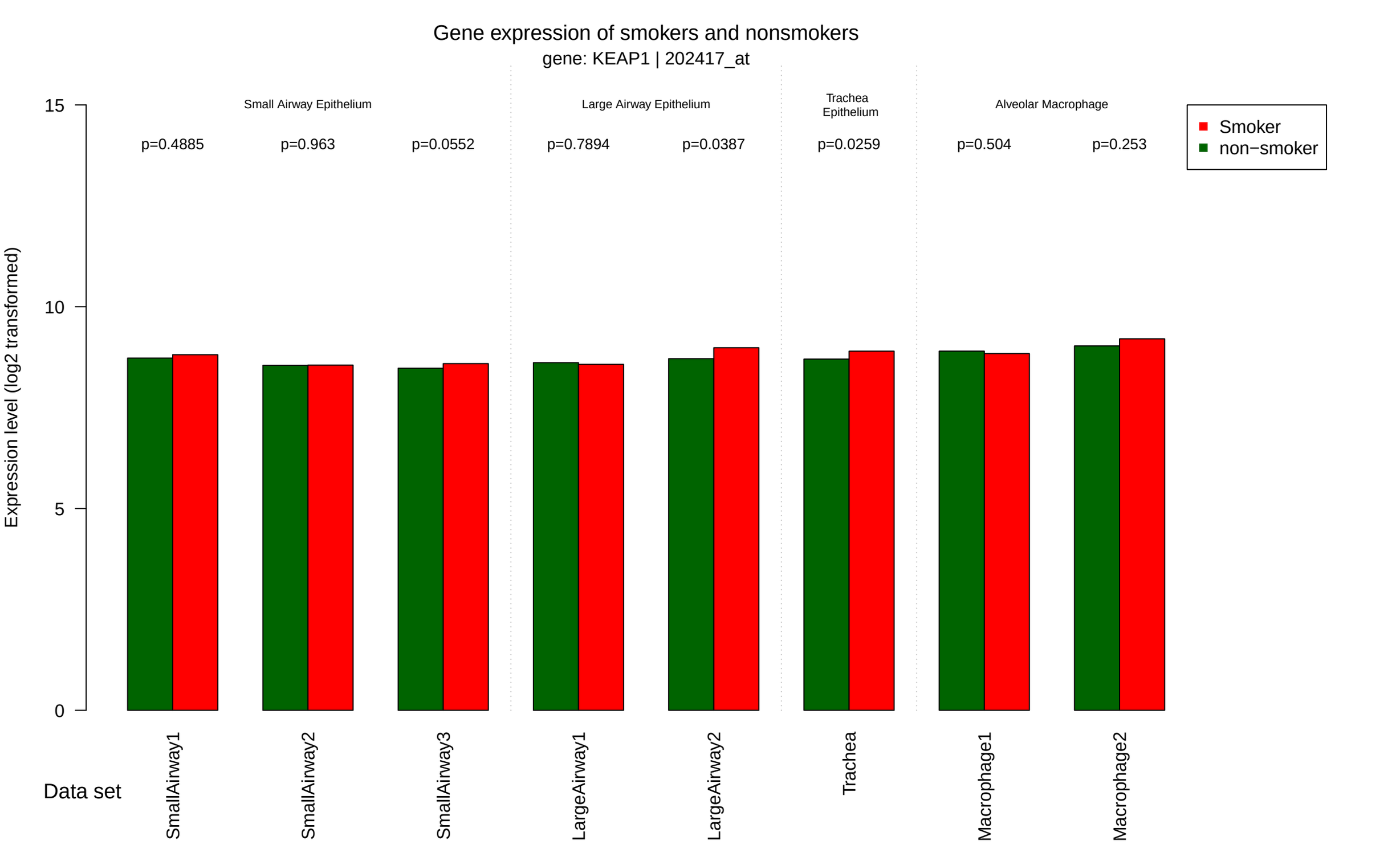
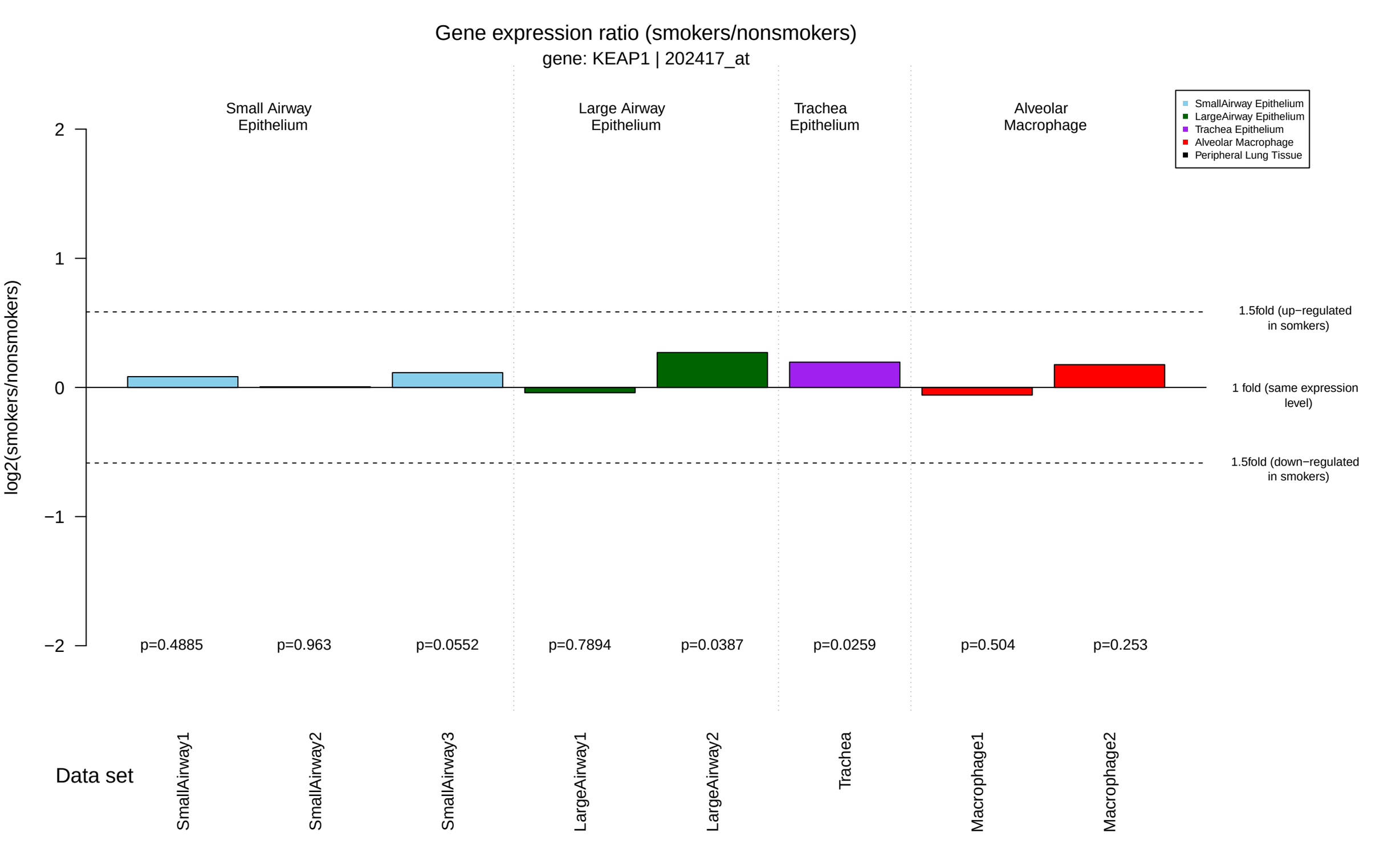
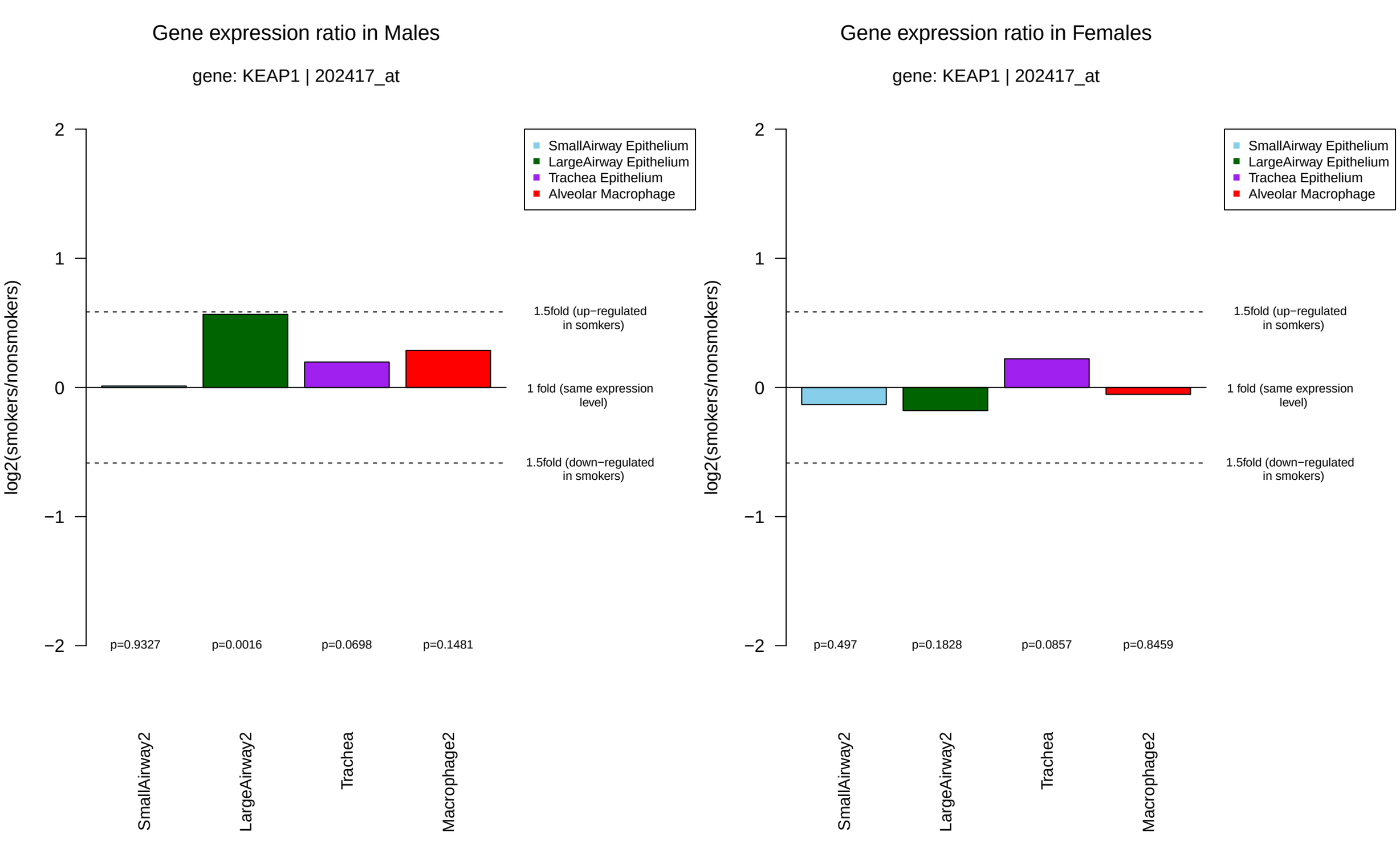
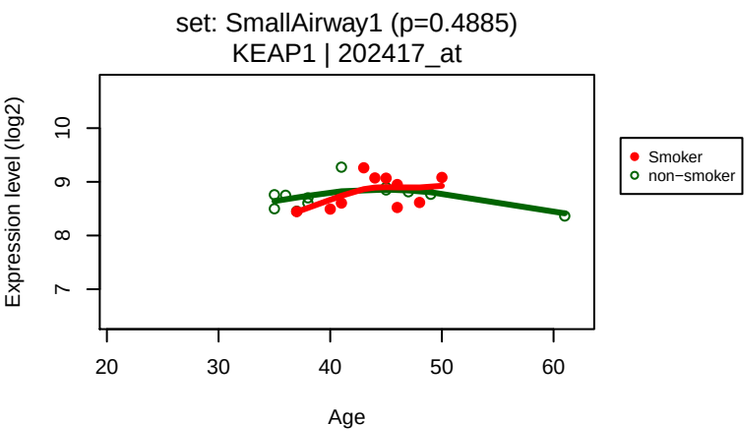
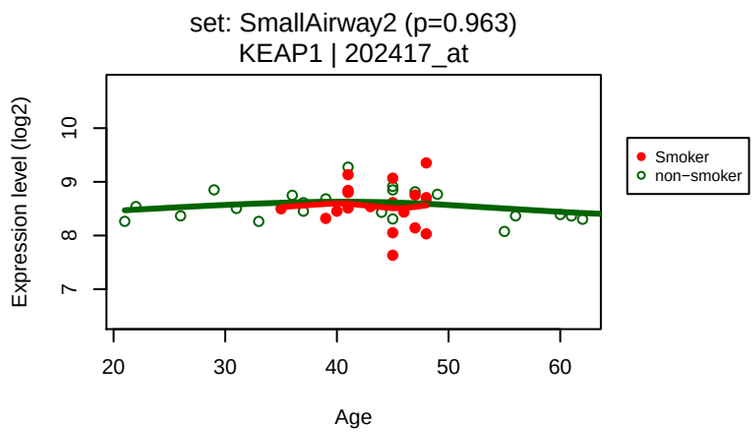
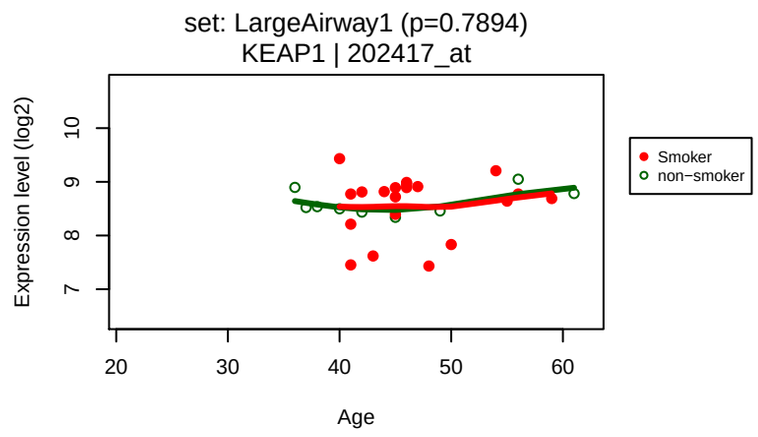

Gene expression of smokers and nonsmokers
gene: KEAP1 | 202417_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4885
p=0.963
p=0.0552
p=0.7894
p=0.0387
p=0.0259
p=0.504
p=0.253
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

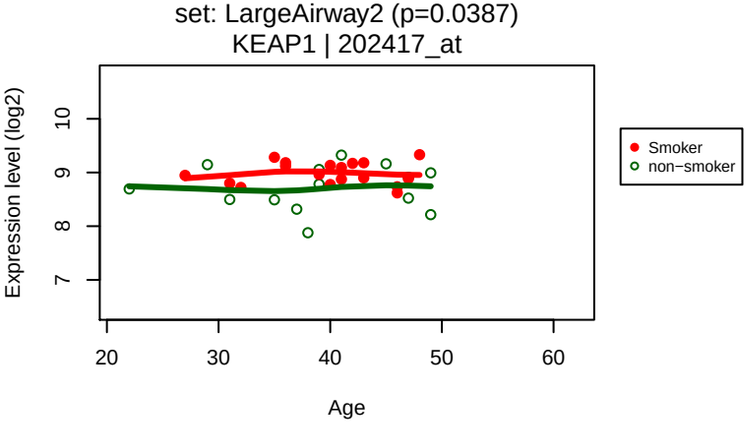

Gene expression ratio (smokers/nonsmokers)
gene: KEAP1 | 202417_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4885
p=0.963
p=0.0552
p=0.7894
p=0.0387
p=0.0259
p=0.504
p=0.253
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

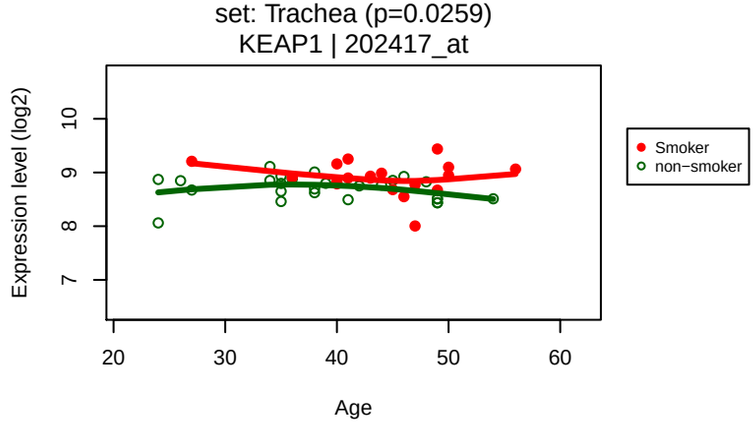

Gene expression ratio in Males
gene: KEAP1 | 202417_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9327
p=0.0016
p=0.0698
p=0.1481
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KEAP1 | 202417_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.497
p=0.1828
p=0.0857
p=0.8459
SmallAirway2
LargeAirway2
Trachea
Macrophage2

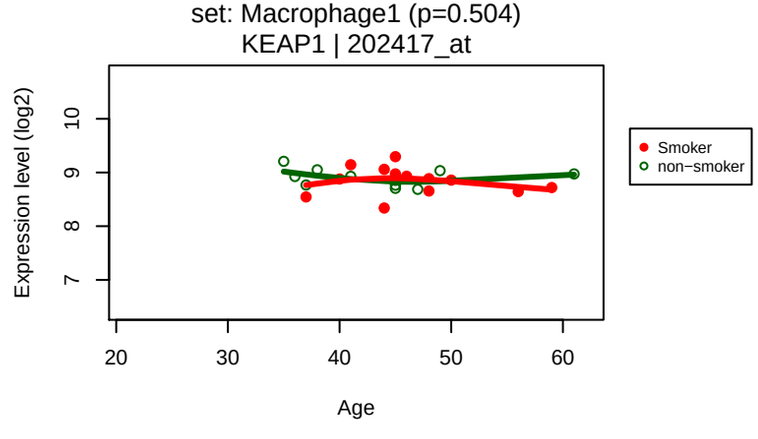

set: SmallAirway1 (p=0.4885)
KEAP1 | 202417_at
Expression level (log2)
Age
Smoker
non−smoker

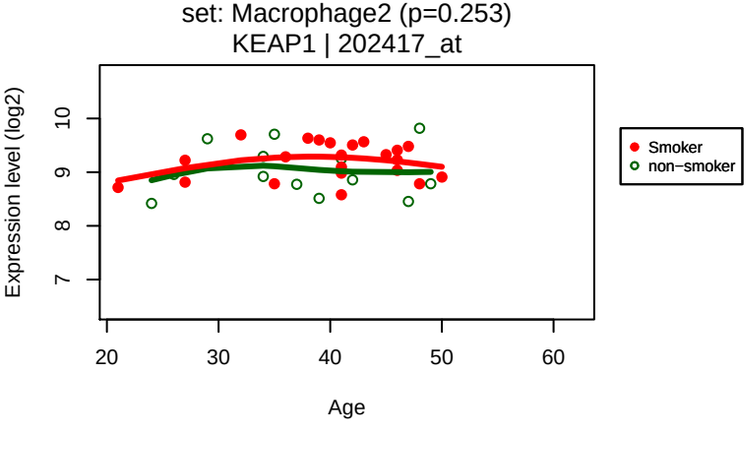

set: SmallAirway2 (p=0.963)
KEAP1 | 202417_at
Expression level (log2)
Age
Smoker
non−smoker

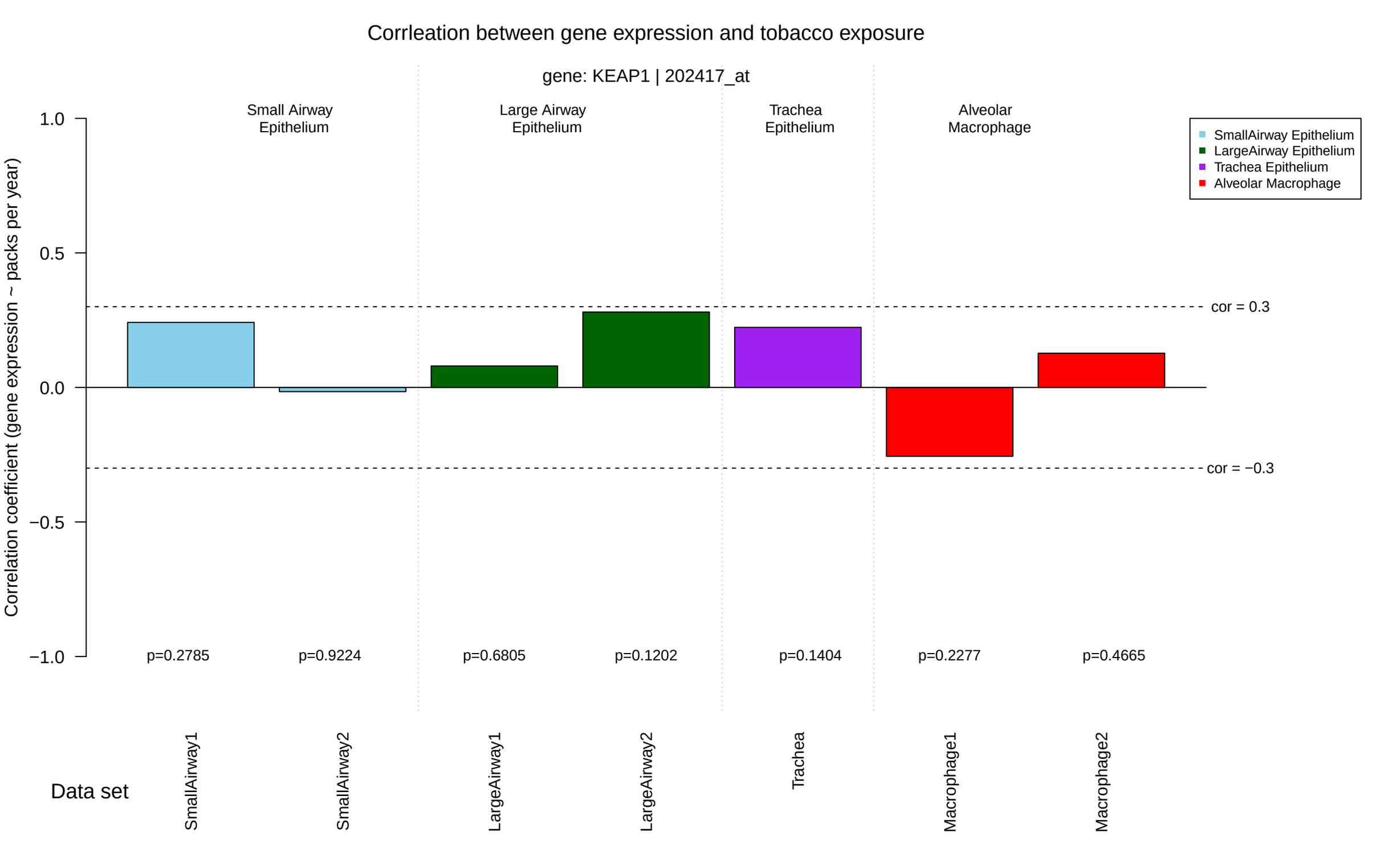

set: LargeAirway1 (p=0.7894)
KEAP1 | 202417_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0387)
KEAP1 | 202417_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0259)
KEAP1 | 202417_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.504)
KEAP1 | 202417_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.253)
KEAP1 | 202417_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KEAP1 | 202417_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2785
p=0.9224
p=0.6805
p=0.1202
p=0.1404
p=0.2277
p=0.4665
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2